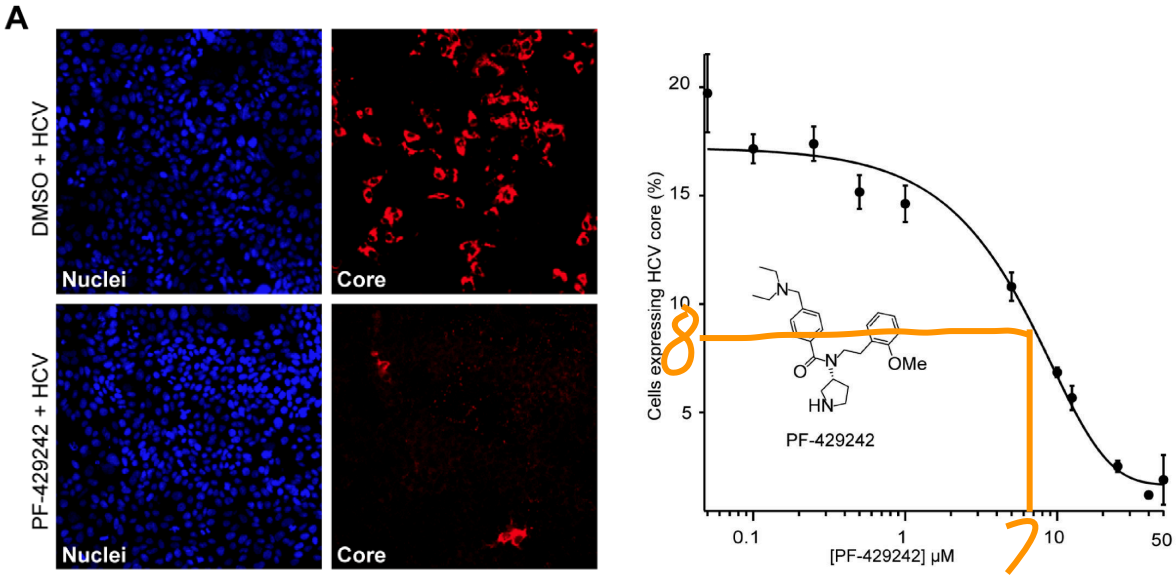
Name _____

Student Number _____

Tutorial 4 Worksheet: Practice Data Analysis Questions

Red is reflection

Identify the Effect of Treatment with Small Molecule X on Virus Infection. Cells derived from hepatocytes (liver cells) were treated with various concentrations (0.01 to 50 µM) of small molecule X (called PF-429242) for 24 hours. Small molecule X was removed, and the cells were then infected with Hepatitis C virus (HCV) for 48 hours. Afterwards, cells were stained with HCV anti-core antibody (**red**) (core protein forms viral nucleocapsid) and Hoechst dye to stain for cell nuclei (**blue**), and then quantified using fluorescence microscopy. (Olmstead et al., 2012)



1. Identify the control(s) in this experiment and briefly explain what they control for. What is the point of staining nuclei?

Control is the DMSO + HCV group, which controls for the effect of the solvent and shows the normal level of HCV infection without the drug.

Staining nuclei helps count the total number and determine the percentage of infected cells. It also helps check whether the drug reduces cell numbers.

My answer was mostly correct, but I missed that DMSO is called a vehicle control.
I also learned that staining nuclei can help check if the drug is toxic to cells, because fewer cells might mean the drug is killing them.

2. Describe the fluorescence images. What conclusions can be drawn?

In the DMSO + HCV group, there are many red signals, which means many cells are infected with HCV. In the PF-429242 + HCV group, there are much fewer reds, but the number of blue nuclei looks similar in both groups.

So that PF-429242 reduces HCV infection without causing a large decrease in cell numbers.

My answer was mostly correct. I should have said that the drug reduces HCV core protein expression, instead of just saying it reduces infection.

3. Based on the graph, estimate what concentration of small molecule X is needed to reduce viral infection by 50%? Draw on the graph how you identified the EC_{50} value.

The EC50 is around 7 μM based on the graph.

My estimate was pretty close. The actual EC50 is around 6.4 μM.
I should be more careful when reading the log scale on the x-axis.

4. Based on the data shown in the fluorescence images, what steps of the viral life cycle could be affected by small molecule X (list all the steps that you think could possibly be blocked by small molecule X)?

PF-429242 could affect viral entry, uncoating, genome replication, viral protein production, or assembly.

I missed attachment in my answer. I also listed too many steps.
The drug may affect early steps like attachment, entry or replication.

5. What experiments could you perform to test whether the entry step is specifically blocked by small molecule X?

We could treat cells with PF-429242 before infection and measure how much virus enters the cells.
Then compare it with untreated cells. If less virus enters the treated cells, it suggests that the drug may block viral entry.

My answer was too simple. I didn't explain how to measure viral entry.
I learned that we can use replication-deficient virus to help check if the drug blocks entry or replication.